Lab 5 Analytical Predictions

Mystery Box Challenge Part 1

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Circuit and phasor conventions

The supplied schematic has a known resistor $R_0 = 1000\,\Omega$ in series with the mystery-box impedance Z. Channel 2 measures the source voltage; Channel 1 measures the voltage across the mystery box. Thus

$$\widetilde{V}_s = \widetilde{V}_{\mathrm{Ch2}}, \qquad \widetilde{V}_z = \widetilde{V}_{\mathrm{Ch1}}, \qquad \widetilde{V}_{1k} = \widetilde{V}_{\mathrm{Ch2}} - \widetilde{V}_{\mathrm{Ch1}}.$$

Use the cosine convention $v(t) = \mathrm{Re}\{\widetilde{V}e^{j\omega t}\}$, with $\omega = 2\pi f > 0$. Phasor magnitudes are **peak amplitudes**, not RMS: $\widetilde{V}_s = 1\angle 0^\circ\,\mathrm{V}$, corresponding to $2\,\mathrm{V_{pp}}$. The components are ideal and their values are positive. Ohm's law gives

$$\widetilde{I} = \frac{\widetilde{V}_s}{R_0 + Z}, \qquad \widetilde{V}_z = \widetilde{V}_s \frac{Z}{R_0 + Z}, \qquad \widetilde{V}_{1k} = \widetilde{V}_s \frac{R_0}{R_0 + Z}. \tag{1}$$

Q3.1.1 Unknown resistor

For a mystery-box resistor R, its impedance is $Z = R$. Therefore

$$\widetilde{I} = \frac{1\,\mathrm{V}}{R_0 + R}, \qquad \widetilde{V}_z = \frac{R}{R_0 + R}\,\mathrm{V}, \qquad \widetilde{V}_{1k} = 1\,\mathrm{V} - \frac{R}{R_0 + R}\,\mathrm{V} = \frac{R_0}{R_0 + R}\,\mathrm{V}.$$

All ratios are positive real numbers, so each voltage has zero phase:

$$\boxed{\widetilde{V}_s = 1\angle 0^\circ\,\mathrm{V}} \qquad \text{Channel 2,}$$

$$\boxed{\widetilde{V}_z = \widetilde{V}_R = \frac{R}{R_0 + R}\angle 0^\circ\,\mathrm{V}} \qquad \text{Channel 1,}$$

$$\boxed{\widetilde{V}_{1k} = \frac{R_0}{R_0 + R}\angle 0^\circ\,\mathrm{V}} \qquad \text{Channel 2 minus Channel 1.}$$

Here $R_0 = 1000\,\Omega$. Both voltages are in phase with the source, and their sum is exactly $1\angle 0^\circ\,\mathrm{V}$.

Q3.1.2 Unknown capacitor

For a mystery-box capacitor C,

$$Z_C = \frac{1}{j\omega C}, \qquad x = \omega R_0 C > 0.$$

Substituting into the voltage divider and multiplying numerator and denominator by $j\omega C$ gives

$$\begin{aligned}\widetilde{V}_z = \widetilde{V}_C &= (1\,\mathrm{V})\frac{1/(j\omega C)}{R_0 + 1/(j\omega C)} = \frac{1}{1+jx}\,\mathrm{V} \\ &= \frac{1-jx}{1+x^2}\,\mathrm{V}.\end{aligned}$$

Using the complex conjugate,

$$|\widetilde{V}_z|^2 = \widetilde{V}_z\widetilde{V}_z^* = \frac{(1-jx)(1+jx)}{(1+x^2)^2}\,\mathrm{V}^2 = \frac{1}{1+x^2}\,\mathrm{V}^2.$$

The real part is positive and the imaginary part is negative. Hence

$$|\widetilde{V}_z| = \frac{1}{\sqrt{1+x^2}}\,\mathrm{V}, \qquad \arg\widetilde{V}_z = -\tan^{-1}x.$$

The required phasors are

$$\boxed{\widetilde{V}_s = 1\angle 0^\circ\,\mathrm{V}}, \qquad \boxed{\widetilde{V}_z = \widetilde{V}_C = \frac{1}{\sqrt{1+(\omega R_0 C)^2}}\angle\Big[-\tan^{-1}(\omega R_0 C)\Big]\ \mathrm{V}}.$$

Extra credit Voltage across the known resistor

Subtract the *complex phasors*, not their magnitudes:

$$\begin{aligned}\widetilde{V}_{1k} &= \widetilde{V}_s - \widetilde{V}_z = \left(1 - \frac{1}{1+jx}\right)\mathrm{V} = \frac{jx}{1+jx}\,\mathrm{V} = \frac{x^2+jx}{1+x^2}\,\mathrm{V}, \\ |\widetilde{V}_{1k}|^2 &= \frac{x^4+x^2}{(1+x^2)^2}\,\mathrm{V}^2 = \frac{x^2}{1+x^2}\,\mathrm{V}^2, \\ \arg\widetilde{V}_{1k} &= 90^\circ - \tan^{-1}x.\end{aligned}$$

Therefore, with $R_0 = 1000\,\Omega$,

$$\boxed{\widetilde{V}_{1k} = \widetilde{V}_R = \frac{\omega R_0 C}{\sqrt{1+(\omega R_0 C)^2}}\angle\Big[90^\circ - \tan^{-1}(\omega R_0 C)\Big]\ \mathrm{V}}.$$

The capacitor voltage lags the source; the known-resistor voltage leads it. Their phases differ by 90°. In these polar answers, inverse tangents are evaluated in degrees. At low frequency, $\widetilde{V}_z \to \widetilde{V}_s$ and $\widetilde{V}_{1k} \to 0$; at high frequency, $\widetilde{V}_z \to 0$ and $\widetilde{V}_{1k} \to \widetilde{V}_s$.

Q3.1.3 Unknown inductor

For a mystery-box inductor L,

$$Z_L = j\omega L, \qquad y = \frac{\omega L}{R_0} > 0.$$

The voltage across the inductor is

$$\begin{aligned}\widetilde{V}_z = \widetilde{V}_L &= (1\,\mathrm{V})\frac{j\omega L}{R_0 + j\omega L} = \frac{jy}{1+jy}\,\mathrm{V} \\ &= \frac{jy(1-jy)}{1+y^2}\,\mathrm{V} = \frac{y^2+jy}{1+y^2}\,\mathrm{V}.\end{aligned}$$

Its magnitude and phase follow from

$$\begin{aligned}|\widetilde{V}_z|^2 &= \frac{(y^2+jy)(y^2-jy)}{(1+y^2)^2}\,\mathrm{V}^2 = \frac{y^2}{1+y^2}\,\mathrm{V}^2, \\ \arg\widetilde{V}_z &= \arg(jy) - \arg(1+jy) = 90^\circ - \tan^{-1} y.\end{aligned}$$

Thus

$$\boxed{\widetilde{V}_s = 1\angle 0^\circ\,\mathrm{V}}, \qquad \boxed{\widetilde{V}_z = \widetilde{V}_L = \frac{\omega L}{\sqrt{R_0^2 + (\omega L)^2}}\angle\left[90^\circ - \tan^{-1}\left(\frac{\omega L}{R_0}\right)\right]\,\mathrm{V}}.$$

The equivalent positive phase is $\tan^{-1}(R_0/(\omega L))$. The worksheet's V_C label in this question is interpreted as V_L, since the assumed component is an inductor.

Extra credit Voltage across the known resistor

$$\begin{aligned}\widetilde{V}_{1k} &= \widetilde{V}_s - \widetilde{V}_z = \left(1 - \frac{jy}{1+jy}\right)\mathrm{V} = \frac{1}{1+jy}\,\mathrm{V} = \frac{1-jy}{1+y^2}\,\mathrm{V}, \\ |\widetilde{V}_{1k}| &= \frac{1}{\sqrt{1+y^2}}\,\mathrm{V}, \qquad \arg\widetilde{V}_{1k} = -\tan^{-1} y.\end{aligned}$$

With $R_0 = 1000\,\Omega$,

$$\boxed{\widetilde{V}_{1k} = \widetilde{V}_R = \frac{R_0}{\sqrt{R_0^2 + (\omega L)^2}}\angle\left[-\tan^{-1}\left(\frac{\omega L}{R_0}\right)\right]\,\mathrm{V}}.$$

The inductor voltage leads the source; the known-resistor voltage lags it. Inverse tangents in the polar answers are in degrees. At low frequency, $\widetilde{V}_z \to 0$ and $\widetilde{V}_{1k} \to \widetilde{V}_s$; at high frequency, $\widetilde{V}_z \to \widetilde{V}_s$ and $\widetilde{V}_{1k} \to 0$.

Consistency check. In all three cases, $\widetilde{V}_z + \widetilde{V}_{1k} = \widetilde{V}_s$. These are analytical predictions only; component identification and experimental verification require measured data.